Mrityunjoy Saha

Ph.D. in Fluid Mechanics

Indian Institute of Technology Jodhpur, India

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RESEARCH INTERESTS

- Hydrodynamic Stability and Transition
- Computational Fluid Dynamics
- Porous Media Flows
- Spectral Methods and Numerical Simulation

EDUCATION

- **Ph.D. in Fluid Mechanics** **2021–2025**
  Indian Institute of Technology Jodhpur, India
  Thesis: Hydrodynamic Stability Analysis on a Class of Complex Flow Systems (Submitted)
  CGPA: 8.69/10
- **M.Sc. in Mathematics (Applied Mathematics)** **2018–2020**
  Cooch Behar Panchanan Barma University, India
  CGPA: 8.19/10
- **B.Sc. (Hons.) in Mathematics** **2015–2018**
  Dinhata College, Cooch Behar Panchanan Barma University, India
  Percentage: 62.5

RESEARCH EXPERIENCE

- **Director's Fellow, IIT Jodhpur** **January 2026–Present**
  Developing Floquet approach-based numerical solvers for stability analysis of time-force induced flow systems, with applications to complex dynamical flow configurations.
- **Senior Research Fellow, IIT Jodhpur** **2024–2025**
  Developed spectral collocation-based numerical solvers for stability analysis of MHD-induced anisotropic porous channel flows and flows in oscillating bounded or semi-bounded domains.
- **Junior Research Fellow, IIT Jodhpur** **2022–2023**
  Conducted theoretical and computational investigations on the hydrodynamic stability of complex flow systems. Developed numerical models to analyze rotational instability mechanisms in wall-bounded porous media flows.

PUBLICATIONS

- **M. Saha** and S. Ghosh, "On the magnetohydrodynamic instability pattern of an anisotropic porous media flow in a rotating channel," *Physics of Fluids*, 38(1), 014122, 2026.
- M. M. Hossain, **M. Saha**, S. Ghosh, and H. Behera, "Stability dynamics of an odd viscosity induced fluid flow over a vibrating inclined surface," *Physics of Fluids*, 37(10), 104108, 2025.
- **M. Saha**, S. Sengupta, S. Mukhopadhyay, and S. Ghosh, "Flow instability in a rotating channel loaded with an anisotropic porous material," *Computers & Fluids*, 299, 106689, 2025.

MANUSCRIPTS

- **M. Saha**, S. Ghosh, M. Roy, and M. Mishra, "Modal and non-modal stability of Poiseuille flow through anisotropic inhomogeneous porous materials." (Under Review)
- **M. Saha**, S. Ghosh, and K. C. Sahoo, "Dual instability in a vertical crossflow-induced oscillating channel flow." (In Preparation)
- S. Panda, **M. Saha**, and S. Ghosh, "Effect of uniform vertical crossflow on rotational instability in a channel flow." (In Preparation)

TEACHING EXPERIENCE

- **Teaching Assistant** **2022–Present**
  Indian Institute of Technology Jodhpur

  Courses assisted: Mathematics I & II, Scientific Computation, Mathematical Modeling, Data Science, Applied Mathematics & Statistics, Abstract Algebra, Linear Algebra, Complex Analysis.

COMPUTATIONAL SKILLS

- **Programming Languages**
  C/C++, Python
- **Scientific Computing & Simulation**
  MATLAB, Mathematica, COMSOL Multiphysics, High Performance Computing (HPC)
- **Domain Expertise**
  Computational Fluid Dynamics (CFD), Hydrodynamic Stability, Porous Media Flows, Numerical PDEs

CONFERENCE PRESENTATIONS

- **78th Annual Meeting of the APS Division of Fluid Dynamics (APS DFD 2025)**
  George R. Brown Convention Center, Houston, Texas, USA.
- **International Conference on Recent Advances in Fluid Mechanics and Nanoelectronics 2023**
  Manipal Institute of Technology, Bengaluru, India.

AWARDS AND SCHOLARSHIPS

- Director's Fellowship, Indian Institute of Technology Jodhpur.
- American Physical Society – Division of Fluid Dynamics Travel Award 2025.
- ANRF International Travel Support for APS DFD 2025.
- CSIR Foreign Travel Grant for APS DFD 2025.
- Qualified Graduate Aptitude Test in Engineering (GATE), Mathematics (2022, 2023).
- DST INSPIRE Scholarship – Government of India.
- SVMCM Scholarship (UG, PG, and Higher Secondary).
- Sitaram Jindal Scholarship for B.Sc.

REFERENCES

- **Dr. Sukhendu Ghosh** (Ph.D. Supervisor)
  Associate Professor
  Department of Mathematics
  Indian Institute of Technology Jodhpur
  Email: sukhendu.math@iitj.ac.in